

Quinn M. Gallagher

Ph.D. Candidate, Princeton University

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EDUCATION

Princeton University

Ph.D. Candidate in Chemical and Biological Engineering

Advisor: Michael A. Webb

Princeton, NJ

Aug. 2022 – Present

University of Pennsylvania

B.S.E. in Chemical and Biomolecular Engineering, *summa cum laude*

Philadelphia, PA

Aug. 2018 – May 2022

HONORS AND AWARDS

Hack Award for Water and the Environment , High Meadows Environmental Institute	2026
AI for Accelerating Invention Graduate Fellow , Princeton University	2025
William R. Schowalter Travel Grant , Princeton University	2025, 2026
Excellence in Teaching Award , Princeton University	2024
Francis Robbins Upton Fellowship , Princeton University	2022
NSF Graduate Research Fellowship , National Science Foundation	2022
Hugh Otto Wolf Memorial Prize , University of Pennsylvania	2022
Future Leader in Chemical Engineering , North Carolina State University	2021
Penn Engineering Exceptional Service Award , University of Pennsylvania	2021
Tau Beta Pi , University of Pennsylvania	2021
Manfred Altman Memorial Award , University of Pennsylvania	2020
Holtz Award , University of Pennsylvania	2019

PUBLICATIONS

- Representing (Complex) Polymers for Machine Learning
S. Jiang, R.A. Patel, **Q.M. Gallagher**, Y. Chen, Q. Zhang, M.A. Webb, *MRS Bulletin* (2026). Submitted.
- Minimizing Data Requirements for Material Property Prediction via Cluster-based Training Set Selection
Q.M. Gallagher, S. Hasko, M.A. Webb, *ChemRxiv* (2026). Revisions submitted at *Nature Computational Science*. [\[Preprint\]](#)
- A Local Structural Basis to Resolve Amorphous Ices
Q.M. Gallagher*, R.J. Szukalo*, N. Giovambattista, P.G. Debenedetti, M.A. Webb, *arXiv* (2026). Revisions submitted at *Nature Communications*. [\[Preprint\]](#) *Equal contribution
- A User's Guide to Your First Self-Driving Liquid Handling Lab
A. Maroulis, D. Waynor, **Q.M. Gallagher**, R.A. Patel, M. Tamasi, D.C. Radford, M.A. Webb, A.J. Gormley, *Digital Discovery* (2026). [\[Link\]](#)
- Pressure-Consistent Iterative Boltzmann Inversion for Coarse-Grained Molecular Dynamics
Z. Yu, R.J. Szukalo, **Q.M. Gallagher**, M.A. Webb, *Journal of Chemical Theory and Computation*, 21, 20 (2025). [\[Link\]](#)
- Sustainable Recovery of Perfluoroalkyl Acids Using a Reusable Molecular Cage
M. Pérez-Ferreiro, **Q.M. Gallagher**, M.A. Webb, A. Criado, J. Mosquera, *Journal of Materials Chemistry A*, 13, 30 (2025). [\[Link\]](#)
- Bayesian Optimization Hackathon for Chemistry and Materials
S. Baird, ..., **Q.M. Gallagher**, ..., Y. Zo, *ChemRxiv* (2025). [\[Preprint\]](#)

2. Data Efficiency of Classification Strategies for Chemical and Materials Design
Q.M. Gallagher, M.A. Webb, *Digital Discovery*, 4, 1 (2025). [\[Link\]](#)
1. Engineering a Surfactant Trap via Postassembly Modification of an Imine Cage
M. Pérez-Ferreiro, **Q.M. Gallagher**, A.B. León, M.A. Webb, A. Criado, J. Mosquera, *Chemistry of Materials*, 36, 18 (2024). [\[Link\]](#)

TEACHING

Princeton University

Guest Lecturer for CBE 512: Machine Learning for Chemical Science and Engineering *October 2025*
Teaching Assistant for CBE 250: Separations in Chemical Engineering *Aug. 2024 – Dec. 2024*
Teaching Assistant for CBE 503: Advanced Thermodynamics *Aug. 2023 – Dec. 2023*
Received the Excellence in Teaching Award from Princeton's School of Engineering and Applied Science.

University of Pennsylvania

Teaching Assistant for CIS 160: Mathematical Foundations of Computer Science *Jan. 2020 – May 2022*
Teacher for Access Engineering *Jan. 2019 – May 2022*

MENTORING

Undergraduate Students:

- Khang Tran, AI Fluency Program (March 2026 –)
- Sonia Hasko, CBE (Sept. 2024 – May 2025)
- David Opong, CBE (Sept. 2023 – May 2024)

High School Students:

- Michelle Gao, Laboratory Learning Program (May 2024 – Aug. 2024)

ACADEMIC SERVICE

Journal Reviewer: *ACS Macro Letters*

Professional Membership: AIChE, ACS

OUTREACH

Graduate Fellow, AI Lab, AI for Accelerating Invention Initiative, Princeton University *Jan. 2026 –*
Reviewer, NSF GRFP Workshop for CBE Students, Princeton University *Sept. 2022 –*
President, Access Engineering, University of Pennsylvania *Sept. 2018 – May 2022*
Vice President, Student Chapter of AIChE, University of Pennsylvania *Sept. 2018 – May 2022*
Secretary, Engineering Deans' Advisory Board, University of Pennsylvania *Sept. 2018 – May 2022*

PRESENTATIONS

14. "Improving Machine Learning Extrapolation for Molecular Discovery" (*Poster*) **ChemAI NYC**, 2026.
13. "Improving Extrapolative Machine Learning for Molecular Design" (*Poster*) **NIST Artificial Intelligence for Materials Science Workshop**, 2026.
12. "Improving Extrapolative Molecular Property Prediction for Screening, Optimization, and Guided Generation" (*Poster*) **CECAM Flagship Workshop**, *Physics-aware Machine Learning for Molecules and Materials*, 2026.
11. "Minimizing Data Requirements for Material Property Prediction *via* Cluster-based Training Set Selection" (*Poster*) **Gordon Research Conference**, *AI for Materials, Energy and Chemical Sciences*, 2026.

10. “Automatic Construction of Data-Driven Collective Variables Using Local Atomic Environments” (*Talk*) **AIChE Annual Meeting**, *CoMSEF: Machine Learning for Soft and Hard Materials II: Hard Matter and Methods*, 2025.
9. “Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships” (*Talk*) **AIChE Annual Meeting**, *CoMSEF: Automated Molecular and Materials Discovery: Integrating Machine Learning, Simulation, and Experiment*, 2025.
8. “Data-driven Order Parameters for Atomic Systems: A Case Study on Amorphous Ices” (*Talk*) **Princeton CBE Graduate Student Symposium**, 2025.
7. “An Interpretable Machine Learning Tool for the Classification of Atomic States in Molecular Simulations” (*Poster*) **6th Molecular Simulations Workshop (NJIT)**, 2025.
6. “Minimizing Data Requirements for Material Property Prediction” (*Poster*) **AutoML Center for Accelerated Formulations Engineering Workshop**, 2025.
5. “Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships” (*Poster*) **Accelerate Conference**, 2025.
4. “Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships” (*Poster*) **New York City AI4Chemistry Summit**, 2025.
3. “Data-driven identification of collective variables for transitions between high-density amorphous and low-density amorphous ice” (*Talk*) **American Chemical Society, Middle Atlantic Regional Meeting**, 2025.
2. “Fewer Experiments, More Knowledge: Data-Efficient Machine Learning for Materials Design” (*Poster*) **Princeton CBE Graduate Student Symposium**, 2024.
1. “Modulating Heterochromatin Formation for Direct Cell Reprogramming” (*Talk*) **North Carolina State University, Future Leaders in Chemical Engineering National Award Symposium**, 2021.